

# Theory-Based Interpretability in Deep Knowledge Tracing via Grounded Transformers

Concepcion Labra <sup>1</sup> , Olga C. Santos <sup>2,\*</sup> 

<sup>1</sup> Department of Artificial Intelligence, Computer Science School, UNED, 28040 Madrid, Spain; clabra@dia.uned.es

<sup>2</sup> PhyUM Research Center, Department of Artificial Intelligence, Computer Science School, UNED, 28040 Madrid, Spain; ocsantos@dia.uned.es

\* Author to whom correspondence should be addressed.

## Abstract

Knowledge Tracing, which enables the estimation of how students' knowledge evolves as they interact with educational content, is a cornerstone of intelligent tutoring systems. Traditional approaches coexist with deep knowledge tracing models that deliver better predictions but suffer from a critical drawback: lack of interpretability, which is particularly problematic in educational contexts. To overcome this limitation, we propose gTransformer, a novel model that unifies the high predictive performance of deep learning with the intrinsic interpretability of traditional approaches like Bayesian Knowledge Tracing. Our design employs an encoder-decoder Transformer that enriches input sequences with parameter estimates from the interpretable model. Attention mechanisms integrate these embeddings into a latent context vector, which is projected into updated parameter values constrained to remain pedagogically grounded to educational constructs while incorporating rich temporal dependencies learned by the model. These enriched parameters then drive predictions through interpretable Bayesian logic. Experiments demonstrating state-of-the-art accuracy show that our model achieves superior diagnostic granularity by identifying student-specific parameters—such as initial knowledge and learning rates—that capture individual longitudinal contexts. By anchoring deep representations to concepts defined by an intrinsically interpretable reference model, gTransformer offers a pedagogically interpretable alternative for data-driven personalization.

**Keywords:** deep knowledge tracing; theory-based interpretability; grounded transformers; bayesian knowledge tracing; educational data analysis; personalized learning

## 1. Introduction

Knowledge Tracing (KT) [1] is a foundational problem in Intelligent Tutoring Systems and Artificial Intelligence in Education. It is formally defined as the supervised sequence modeling task of estimating a student's evolving mastery based on their historical interactions. The input to the system consists of a temporal sequence  $\mathcal{S} = \{(q_1, r_1), (q_2, r_2), \dots, (q_{t-1}, r_{t-1})\}$ , where each tuple represents a question (or skill)  $q_i$  and the student's binary response  $r_i \in \{0, 1\}$ . The task objective is to predict the output  $\hat{y}_t = P(r_t = 1 | q_t, \mathcal{S})$ , signifying the probability of a correct response to the current question  $q_t$  given the observed history. By monitoring these longitudinal estimations, we can track the dynamic evolution of student knowledge, providing a data-driven foundation for personalized instruction and targeted pedagogical interventions. In an era where educational environments are becoming increasingly digital and diverse, the ability to accurately

Received:

Revised:

Accepted:

Published:

**Copyright:** © 2026 by the authors.

Submitted to *Journal Not Specified* for possible open access publication

under the terms and conditions of the

[Creative Commons Attribution](#)

(CC BY) license.

track and interpret student mastery is not merely a technical optimization but a critical requirement for effective, scalable, and learner-centred education.

Historically, the development of KT has been characterized by a dichotomy between two distinct paradigms [2]. The first, which predominated in the field for years, is characterized by probabilistic models such as Bayesian Knowledge Tracing [1] and Factor Analysis Models based on logistic regression—including Additive Factor Models [3], Performance Factor Analysis [4], and Knowledge Tracing Machines [5]. Factor Analysis Models are theoretically supported by the Item Response Theory (IRT) [6], which has played a large role in educational assessment and measurement. These traditional approaches are intrinsically interpretable: they model learning as a process governed by explicit parameters related to pedagogical, cognitive or psychometric principles. Because the model structure mirrors a theoretical understanding of human learning, educators can inspect these values to validate the model’s reasoning and trust its outputs. However, this transparency comes at the cost of representational rigidity; the simplifying assumptions required by these models often limit their predictive power, causing them to struggle with the complex, non-linear, and long-range dependencies inherent in real-world student behavior [7].

The second paradigm emerged with the advent of Deep Knowledge Tracing (DKT) [8], which leverages deep learning techniques ranging from initial Recurrent Neural Networks to more recent Transformer architectures [9]. The latter, based on encoder-decoder components and attention mechanisms, are especially relevant as they provide the foundation for current Large Language Models. By treating student interaction histories as temporal sequences, Transformer models apply techniques similar to those that characterize LLMs. DKT models have achieved state-of-the-art predictive performance, significantly outperforming classical baselines due to a structural high capacity that allow them to extract latent features from vast amounts of data [2]. However, this substantial gain in predictive performance has come at the cost of model transparency, creating a significant interpretability gap. Deep learning models are notoriously opaque “black boxes”, where the internal high-dimensional representations bear no direct correspondence to constructs that are interpretable in terms of human-understandable concepts or established domain knowledge. In a high-stakes domain such as education—where algorithmic decisions influence learning trajectories, grades, and opportunities—this lack of transparency poses critical challenges for practical adoption.

Extensive research has sought to bridge this gap, with Bai et al. [10] classifying modalities into *post-hoc* and *ante-hoc* categories. While *post-hoc* methods aim to extract explanations from black-box models after training, *ante-hoc* approaches seek to achieve intrinsic interpretability by design, ensuring the model’s internal logic is transparent from the outset. Within this latter category, recent models have attempted to achieve intrinsic transparency by integrating attention mechanisms [11,12] or by incorporating side information such as Item Response Theory (IRT) parameters [13–15]. These efforts aim to make the decision-making process visible, moving away from pure black-box architectures toward systems whose internal states can be probed or visualized.

However, from the perspective of domain experts such as teachers and pedagogical designers, these solutions frequently fall short. Post-hoc explanations often rely on technical relevance scores or local approximations that remain disconnected from established pedagogical principles [16] while most current ante-hoc methods do not produce outputs that map directly to human-understandable educational constructs. As noted in various surveys [10,17], when deep learning models incorporate domain knowledge, the primary objective uses to be the optimization of predictive performance, while interpretability is often overlooked or left as an unaddressed concern.

Consequently, there is a critical need for new approaches that move beyond technical transparency toward true pedagogical interpretability. For a Knowledge Tracing system to be confidently adopted in high-stakes educational environments, its estimations must be grounded in trusted and empirically validated principles derived from relevant fields such as pedagogy, cognitive science, or psychometrics. Educators require models that speak their language—tracking mastery not through opaque latent features, but through concepts explicitly defined by established frameworks, including the skills, prior knowledge, or learning rates postulated by traditional approaches such as BKT and Factor Analysis Models.

To address this challenge, we introduce the *grounded Transformer (gTransformer)*, a variant of attention-based Transformers that implements a novel *Latent Grounding* methodology to bridge the gap between deep learning expressiveness and theoretical alignment. This approach explicitly anchors the Transformer’s latent representations to the semantic constructs of an intrinsically interpretable reference model. The gTransformer architecture utilizes an encoder-decoder design with attention mechanisms to create context vectors that encode complex student interaction patterns. These vectors are projected into enriched parameters, which then drive predictions using the reference model’s interpretable logic. By doing so, the model’s estimations can be tracked down to domain knowledge and theoretical principles. This architecture satisfies the requirements for intrinsic interpretability by providing “grounded interpretability” by design, as the estimations are expressed through the same well-validated concepts used in established interpretable systems.

Our approach contributes to bridging the dichotomy between high-capacity deep learning and domain-specific theoretical transparency, providing two practical advantages. First, compared to non-interpretable DKT models, gTransformer offers intrinsic pedagogical transparency by generating actionable parameters—such as prior knowledge and learning rates—that are directly meaningful to educators. Second, compared to traditional approaches, gTransformer is context-aware, i.e it is able to encode individual patterns taking into account the history of student interactions, overcoming the limitations of rigid population-level parameters and Markovian assumptions characteristic of BKT. For instance, it can distinguish between a learner requiring foundational remediation and one exhibiting high acquisition momentum even when their immediate local performance appears indistinguishable. By integrating these capabilities, gTransformer provides a foundation for truly individualized student-centered instruction, allowing educators to move beyond predicting *if* a student will succeed toward understanding *why* they are struggling.

We evaluate our approach across four primary dimensions. First, we demonstrate parameter recovery accuracy by computing Pearson correlations and generating scatter plot visualizations between the model’s predicted parameters and theoretical targets from a fitted BKT model, achieving correlations exceeding 0.7 for both initial mastery and learning rate estimates. Second, we conduct ablation studies to quantify the interpretability cost, showing that theory-guided grounding achieves full interpretability with less than 0.5 percentage points of predictive accuracy loss. Third, we employ a dual evaluation protocol that separately measures both standard supervised predictions and interpretable predictions, demonstrating that the latter outperform classical BKT AUC by 10.8% while remaining highly competitive with non-interpretable DKT baselines. Finally, we present case study visualizations of learning trajectories across distinct learning situations characterized by different combinations of student prior knowledge and learning rates. These analyses show how gTransformer’s predictions are context-aware, effectively incorporating the history of interactions in a way that Markovian models like BKT are incapable of. By capturing these behavioral nuances, the model provides longitudinal, individualized diagnostics

that support pedagogical decisions such as appropriate placement, pacing, and targeted intervention.

Our research is guided by three primary research questions:

- **RQ1: Theory-Based Interpretability Through Grounded Transformers**  
Can deep knowledge tracing models achieve state-of-the-art predictive performance while providing interpretability grounded in established principles and theories? Specifically, can we design a transformer architecture that produces pedagogically meaningful mastery estimations that are explainable through a causal and interpretable logic, such as Bayesian Knowledge Tracing?
- **RQ2: Trade-Offs Between Predictive Performance and Interpretability**  
How do the metrics of the supervised, interpretable, and BKT predictions compare? What is the cost of interpretability in terms of AUC? How much predictive gain do the interpretable grounded predictions achieve compared to traditional BKT?
- **RQ3: Practical Value for Student-Centered Personalization**  
Beyond providing interpretable diagnostics, does the high capacity of gTransformer to capture intricate interaction patterns offer advantages over traditional models? Specifically, can these capabilities be leveraged to enhance student-centered personalization relative to population-based models such as Bayesian Knowledge Tracing?

The remainder of this paper is structured as follows. Section 2 reviews related work, while Section 3 describes the gTransformer model, detailing the construction of embeddings, the grounding mechanism, and the prediction logic. Section 4 presents the experimental setup, followed by a discussion of the empirical results in Section 5. Finally, Section 6 provides the concluding remarks.

## 2. Related Work

### 2.1. Bayesian Knowledge Tracing

For this study, we selected BKT [1] as our reference theoretical framework. It models the learning process as a hidden Markov process governed by four parameters providing a causal narrative of student performance that is intuitive to educators and aligned with cognitive science principles:

- **Initial Mastery:** The probability a student knows a skill prior to practice;
- **Learn Rate:** The probability of transitioning from unlearned to learned states after a practice opportunity;
- **Guess:** The probability of a correct response despite a lack of mastery; and
- **Slip:** The probability of an incorrect response despite mastery.

Standard BKT suffers from a significant limitation: it typically operates at a population level, estimating a single set of parameters for all students interacting with a given skill. While *Individualized BKT* approaches exist, they are often computationally expensive and struggle with data sparsity [18]. The Representational Grounding used in the gTransformer model offers a new way to address individualization by leveraging the capabilities of Transformers to compress student interaction history into a context vector. This allows the model to infer personalized initial knowledge and learning rates dynamically, transforming population-level BKT insights into high-granularity student diagnostics.

### 2.2. Deep Knowledge Tracing

The application of deep learning to knowledge tracing was initiated by the Deep Knowledge Tracing (DKT) model [8], which framed student performance as a sequence modeling task. While pioneering, DKT was constrained by the inherent limitations of Recurrent Neural Networks (RNNs). These architectures are difficult to train [19] and their

sequential nature precludes parallelization within training examples, a bottleneck that becomes critical at longer sequence lengths where memory constraints limit efficiency[9].

The introduction of attention-based architectures, such as Self-Attentive Knowledge Tracing (SAKT) [11] and Separated Self-Attentive Neural Knowledge Tracing (SAINT) [20], marked a significant shift toward the attention-based Transformers that characterizes modern Large Language Models. They leverage this architecture ability to capture long-range dependencies, allowing for a non-Markovian representation of a student's entire history. More recently, Attentive Knowledge Tracing (AKT) [12] further refined this by incorporating skill difficulty and context-aware attention mechanisms that weigh past interactions based on their temporal distance.

Beyond the prominent recurrent and attention-based paradigms, the research community has explored several other architectural directions to capture the multifaceted nature of student learning. Notable categories include Memory-Augmented models that utilize external storage to explicitly track concept mastery, Graph-Based approaches that model the topological relationships between skills, and specialized variations such as Text-Aware or Forgetting-Aware models that incorporate exercise semantics and temporal decay into the tracing process [2]. While these innovations have further enhanced predictive accuracy, they share the common challenge of balancing model complexity with the pedagogical transparency required for meaningful educational intervention.

### 2.3. Interpretability in Deep Knowledge Tracing

Interpretability in DKT remains an open challenge despite a variety of attempts, which can be categorized into two primary modalities: *post-hoc* and *ante-hoc*. Post-hoc methods attempt to explain trained black-box models by utilizing either model-specific techniques, such as Layer-Wise Relevance Propagation (LRP) [21], or model-agnostic approaches like Local Interpretable Model-agnostic Explanations (LIME) [22]. However, Rudin [16] argues that relying on post-hoc explanations for high-stakes decision-making is fundamentally flawed. For teachers and domain experts, these methods offer limited utility as they produce explanations tied to technical model structures rather than pedagogically sound principles, necessitating significant technical expertise for meaningful interpretation.

The second category, *ante-hoc* methods, pursues intrinsic interpretability through two primary approaches. The first consists of traditional models that rely on transparent structures, such as BKT or Factor Analysis Models. The second encompasses DKT variants that achieve interpretability by incorporating additional modules or utilizing side information [10]. Among these, attention mechanisms have emerged as the most prominent architectural extensions. They have been integrated into a wide range of models, from the pioneering Self-Attentive Knowledge Tracing (SAKT) [11] to more recent architectures like Attentive Knowledge Tracing (AKT) [12], which achieves high predictive performance by utilizing multi-scale attention to capture interaction patterns across varying temporal resolutions. However, relying solely on attention weights presents similar practical obstacles to those of post-hoc methods for educational practitioners, as these mechanisms do not provide an explicit interpretation grounded in pedagogically sound principles.

The second approach to ante-hoc interpretability involves the integration of side information, with Item Response Theory (IRT) [6] serving as the most prominent foundation. IRT estimates the probability of a correct response based on a student's latent ability and the item's difficulty. This theory has been coupled with DKT models such as Deep-IRT [13], which utilizes dynamic key-value memory networks to capture learners' trajectories and uses inferred abilities and difficulties to predict performance. Similarly, TC-MIRT [14] embeds multidimensional IRT parameters to predict student states and generate interpretable parameters. Other relevant models include KIKT [23], which harnesses IRT



to simulate student performance; LANA [24], which adopts an IRT variant for ability clustering; and QIKT [15], which features question-centric representations integrated with an interpretable IRT layer. Beyond IRT, researchers have leveraged diverse principles, such as constructive learning [25,26], learning and forgetting curves [27], finite state automata (FSAs) [28], classical test theory (CTT) [29], and monotonicity constraints [30].

In contrast to these models, whose diagnostic utility is often restricted by the specificity and scope of the single underlying theory serving as side information [10], gTransformer offers a more flexible framework. It can be grounded in any established theoretical reference, effectively decoupling the expressiveness of deep learning from the constraints of specific pedagogical models.

## 2.4. Informed Machine Learning

The integration of domain knowledge into deep learning has been broadly explored through conceptual frameworks such as Theory-Guided Data Science (TGDS) [31], Physics-Informed Machine Learning (PIML) [32], and Informed Machine Learning (IML) [17].

TGDS emerged as a paradigm to address the limitations of purely data-driven “black-box” models, which may produce results inconsistent with established domain knowledge [31]. By explicitly requiring scientific consistency, TGDS integrates theoretical insights into the learning process to ensure that models remain scientifically plausible. This integration is typically achieved via theory-guided learning—where domain-specific invariants serve as regularization terms—or through the design of neural architectures that respect the structural constraints of the problem [31,33]. A prominent implementation of this paradigm is Physics-Informed Neural Networks (PINNs) [32,34], which embed differential equations directly into the loss function, forcing the model to satisfy theoretical constraints while learning from empirical data.

A widely adopted technique to enforce this consistency involves incorporating domain-specific constraints into the loss function as follows [31,35]:

$$\mathcal{L} = \mathcal{L}_{TRN}(Y_{true}, Y_{pred}) + \gamma \mathcal{L}_{THEORY}(Y_{pred}), \quad (1)$$

where  $\mathcal{L}_{TRN}$  measures the supervised error between true labels  $Y_{true}$  and predicted labels  $Y_{pred}$ , while  $\mathcal{L}_{THEORY}$  penalizes violations of theoretical constraints, weighted by the hyperparameter  $\gamma$ .

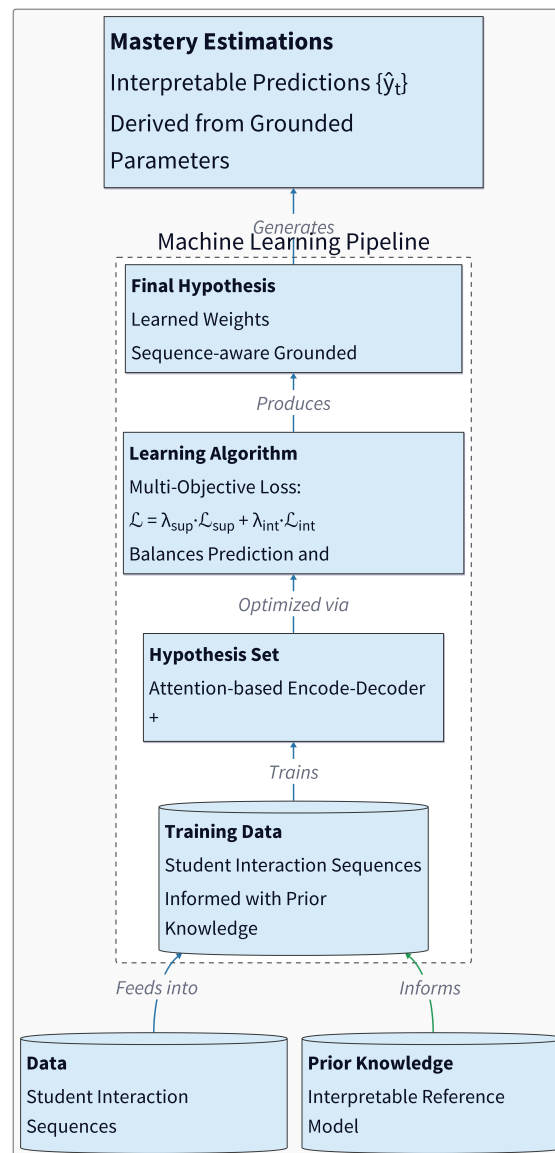
Von Rueden et al. [17] formalized these efforts into an IML taxonomy, which categorizes knowledge integration by its source, representation, and the stage of the machine learning pipeline where it is incorporated. Knowledge can be represented as algebraic equations, logical rules, or probabilistic priors, and integrated at various stages: within the training data, through the design of tailored architectures, or by influencing the model’s final output.

We operationalized these taxonomic dimensions in gTransformer by integrating theoretical priors from an interpretable BKT model into the input layer and employing a multi-objective loss function to anchor the model’s predictions to BKT parameters. However, while most IML approaches focus on enhancing predictive performance [17], gTransformer aims to achieve theory-based interpretability by grounding estimations in sound pedagogical concepts, as detailed in Section 3.1.

## 3. The gTransformer Model

### 3.1. Dimensions of Informed Machine Learning in gTransformer

Figure 1 illustrates the information flow through a gTransformer, mapped to the taxonomic dimensions of IML as defined by Von Rueden et al. [17].



**Figure 1.** Dimensions of Informed Machine Learning in gTransformer.

Within the context of our architecture, these dimensions are operationalized as follows:

- **Prior Knowledge:** The source of theoretical information comprises simulation results from a reference model, typically expressed as parameter values that represent foundational pedagogical constructs.
- **Training Data:** Empirical student interaction histories are augmented with theoretical priors to guide the learning process toward pedagogically valid states.
- **Hypothesis Set:** An attention-based Transformer architecture is employed to provide the inductive biases necessary for grounding internal latent representations within the conceptual framework of the reference model.
- **Learning Algorithm:** The model is optimized via a multi-objective loss function formulated to balance the trade-off between predictive performance and pedagogical interpretability.
- **Final Hypothesis:** A set of learned weights that drive latent representations from which parameter values aligned with the reference model can be extracted.

### 3.2. Architecture

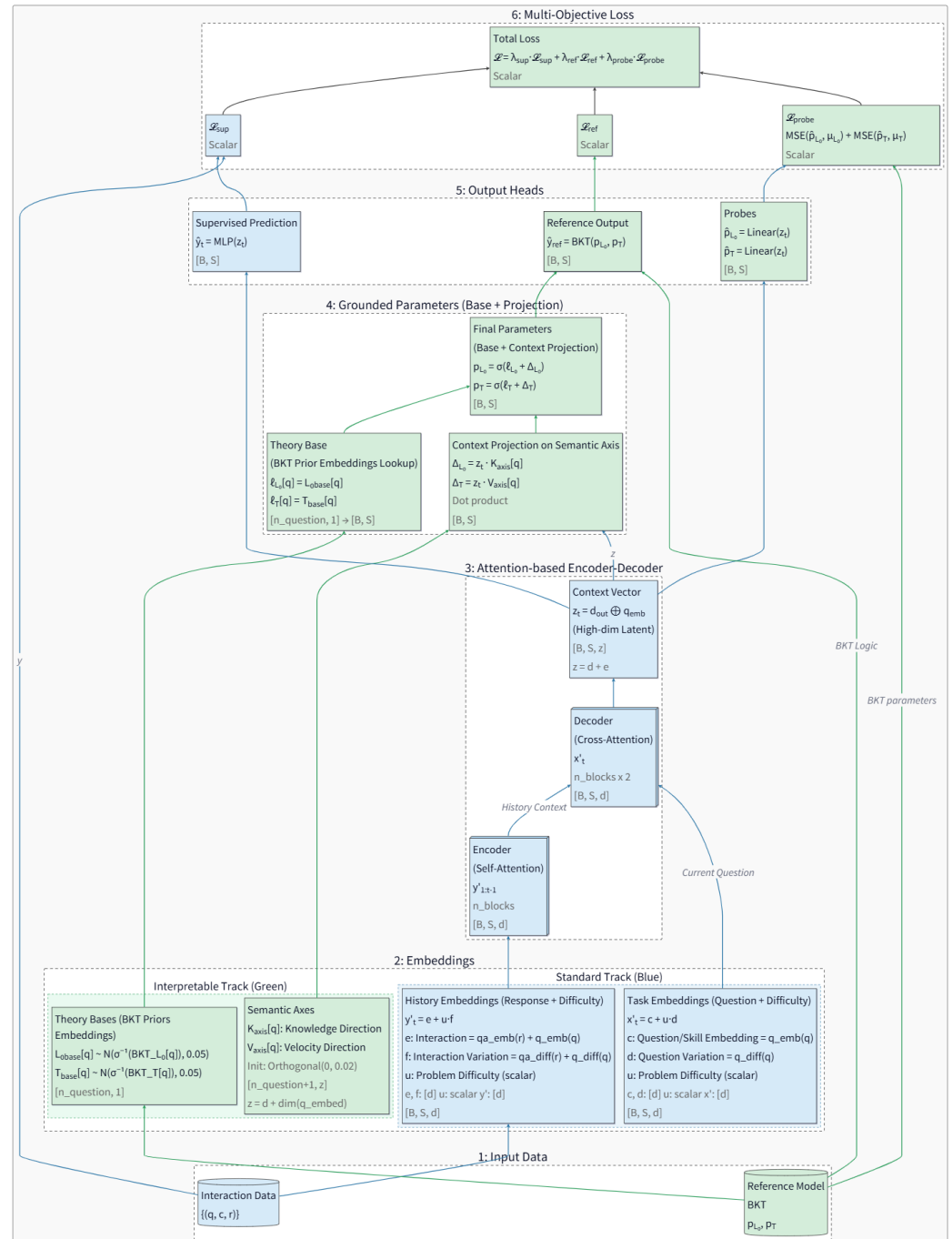
The gTransformer architecture, illustrated in Figure 2, implements a theory-guided deep learning framework that integrates BKT concepts throughout the processing pipeline. The architecture is organized into six functional layers, which are detailed in the following subsections. Specifically, we: (i) begin with input data comprising student interaction sequences along with prior knowledge from the Bayesian reference model (Section 3.2.1); (ii) proceed to difficulty-aware embeddings that encode skill-specific variations (Section 3.2.2); (iii) describe the attention-based encoder-decoder mechanism used to construct contextualized latent representations (Section 3.2.3); (iv) explain how grounded parameters are derived through additive composition in logit space, combining theoretical bases with contextual projections onto semantic axes (Section 3.2.4); (v) present the three parallel output heads intended to generate supervised predictions, interpretable predictions, and representation probes (??); and (vi) conclude with the multi-objective loss function that orchestrates the training process to achieve both predictive performance and theory-based interpretability (??).

#### 3.2.1. Input Components

The data flow begins with the ingestion of student interaction sequences enriched with population-level parameters from a pre-fit BKT reference model. Student interactions are represented as temporal sequences of tuples  $(q_t, c_t, r_t)$ , where  $q_t$  denotes the question identifier,  $c_t$  represents the associated knowledge component, concept or skill, and  $r_t \in \{0, 1\}$  indicates the binary correctness of the student's response at timestep  $t$ .

These empirical interaction traces are augmented with theoretical priors extracted from the BKT reference model. For each knowledge component  $c$ , we obtain population-level estimates of four fundamental BKT parameters: prior knowledge  $P(L_0)_c$  (the probability that a randomly selected student has already mastered concept  $c$  before any practice), learning rate  $P(T)_c$  (the probability of transitioning from non-mastery to mastery after a single learning opportunity), guess probability  $P(G)_c$  (the likelihood of answering correctly despite not having mastered the concept), and slip probability  $P(S)_c$  (the likelihood of answering incorrectly despite having mastered the concept). These parameters serve as the theoretical anchor that grounds the model's internal representations, ensuring that learned latent features remain pedagogically interpretable throughout the deep learning pipeline.





**Figure 2.** Functional layers of the gTransformer Architecture (blue components denote standard Transformer elements; green components denote gTransformer-specific theoretical grounding): (1) *Input Components*: student interaction data enriched with BKT reference parameters; (2) *Difficulty-Aware Embeddings*: task-specific embeddings ( $x'$ ) composed of concept bases and difficulty variations, and interaction history embeddings ( $y'$ ) encoding prior responses; (3) *Attention-based Encoder-Decoder*: a stacked Transformer architecture with  $n$  encoder blocks (self-attention on interaction history) and  $2n$  decoder blocks (alternating self-attention on current questions and cross-attention with encoded history), producing a high-dimensional context vector  $z_t$  by concatenating the final decoder output  $d_{out}$  with question embeddings; (4) *Grounded Parameters*: additive composition of BKT-initialized theoretical bases ( $\ell_{L_0}, \ell_T$ ) and contextual projections ( $\Delta_{L_0}, \Delta_T$ ) computed via dot products of  $z_t$  with learnable semantic axes ( $K_{axis}, V_{axis}$ ), yielding pedagogically-grounded parameters ( $p_{L_0}, p_T$ ); (5) *Output Heads*: supervised prediction ( $\hat{y}_{sup}$ ) from an MLP applied to  $z_t$ , interpretable prediction ( $\hat{y}_{ref}$ ) generated by feeding enriched parameters (projected from  $z_t$ ) through BKT logic to preserve causal explainability (Reference Output), and linear probes ( $\hat{p}_{L_0}, \hat{p}_T$ ) for internal representation analysis; finally (6) *Multi-Objective Loss*: weighted combination of supervised loss ( $\mathcal{L}_{sup}$ ), reference output loss ( $\mathcal{L}_{ref}$ ), and probing loss ( $\mathcal{L}_{probe}$ ) for aligning neural representations with pedagogical theory.

### 3.2.2. Difficulty-Aware Embeddings

Rather than treating questions and responses as discrete atomic identifiers, gTransformer decomposes them into structured embedding vectors that capture both semantic identity and task-specific variation. This decomposition follows a Rasch-inspired additive composition principle, where each embedding is constructed as the sum of a base representation and a difficulty-modulated perturbation.

For question embeddings, we define:

$$x'_t = c_q + u_q \cdot d_c \quad (2)$$

where  $c_q \in \mathbb{R}^d$  represents the invariant concept base embedding (encoding the semantic identity of the knowledge component),  $u_q \in \mathbb{R}$  is a scalar difficulty parameter specific to question  $q$ , and  $d_c \in \mathbb{R}^d$  is the concept-specific difficulty variation axis. This formulation allows multiple questions targeting the same concept to share a common semantic core ( $c_q$ ) while differing in their position along a learned difficulty direction ( $d_c$ ), with the magnitude of displacement controlled by the per-question difficulty scalar ( $u_q$ ).

Similarly, for interaction history embeddings (encoding both the question and the student's response), we employ:

$$y'_t = e_{c,r} + u_q \cdot f_{c,r} \quad (3)$$

where  $e_{c,r} \in \mathbb{R}^d$  is the interaction base embedding (jointly encoding the concept  $c$  and response outcome  $r$ ), and  $f_{c,r} \in \mathbb{R}^d$  is the interaction-specific difficulty variation axis. These embeddings serve as the primary inputs to the Transformer's attention mechanism, providing multi-dimensional representations that enable attention-driven discovery of relevant behavioral patterns across the student's learning history.

### 3.2.3. Attention-Based Encoder-Decoder

The core processing engine of gTransformer is a stacked Transformer architecture composed of  $n$  encoder blocks followed by  $2n$  decoder blocks. The encoder processes the student's longitudinal interaction history through  $n$  sequential self-attention layers, transforming the sequence of interaction embeddings  $\{y'_1, y'_2, \dots, y'_{t-1}\}$  into a tensor of contextualized latent representations  $\mathbf{Y}_{\text{enc}} \in \mathbb{R}^{(t-1) \times d}$ . Each encoder block applies multi-head self-attention with monotonic causal masking to prevent information leakage from future timesteps, followed by position-wise feed-forward networks and residual connections with layer normalization.

The decoder architecture employs a distinctive alternating pattern of  $2n$  sequential layers. Odd-numbered decoder layers (layers 1, 3, 5,  $\dots$ ,  $2n - 1$ ) perform self-attention over the current question embeddings  $x'_t$ , allowing the model to examine multiple aspects of the current task without incorporating response history. Even-numbered decoder layers (layers 2, 4, 6,  $\dots$ ,  $2n$ ) perform cross-attention between the current question representation and the encoder's output  $\mathbf{Y}_{\text{enc}}$ , retrieving relevant knowledge from the student's encoded learning trajectory. Critically, only the even-numbered cross-attention layers include position-wise feed-forward networks, while odd-numbered self-attention layers omit them. This asymmetric design creates  $n$  complete encoder-decoder units, where each unit consists of a self-attention layer (to process current task features) paired with a cross-attention layer (to integrate historical context).

The final decoder layer produces an output tensor  $\mathbf{d}_{\text{out}} \in \mathbb{R}^{t \times d}$ , which is concatenated with the question embedding to form the high-dimensional context vector:

$$z_t = [\mathbf{d}_{\text{out},t}; x'_t] \in \mathbb{R}^z \quad (4)$$

where  $z = d + e$  represents the combined dimensionality of the decoder output and question embedding spaces. This context vector serves as a comprehensive latent encoding of the student's current cognitive state, synthesizing information from both the observed interaction history and the current task characteristics.

### 3.2.4. Grounded Parameters

To ensure pedagogical interpretability, gTransformer estimates BKT parameters through an additive composition mechanism operating in logit space. For each knowledge component  $c$ , we define theoretical base parameters  $\ell_{L_0,c}$  and  $\ell_{T,c}$ , initialized as the logit-transformed population-level BKT priors:

$$\ell_{L_0,c} = \text{logit}(P(L_0)_c), \quad \ell_{T,c} = \text{logit}(P(T)_c) \quad (5)$$

These bases are implemented as learnable scalar embeddings with Gaussian initialization centered at the theoretical values ( $\mathcal{N}(\mu = \text{logit}(P), \sigma = 0.05)$ ), ensuring that the model begins training from pedagogically valid priors while allowing fine-tuning through gradient descent.

Contextual adjustments are computed via dot products between the latent context vector  $z_t$  and learnable semantic axes. For each concept  $c$ , we define a knowledge axis  $\mathbf{K}_c \in \mathbb{R}^z$  and a learning velocity axis  $\mathbf{V}_c \in \mathbb{R}^z$ . The contextual projections, representing the student-specific Knowledge Gap ( $k_s$ ) and Learning Velocity ( $v_s$ ), are calculated as:

$$k_{s,t} = z_t \cdot \mathbf{K}_c, \quad v_{s,t} = z_t \cdot \mathbf{V}_c \quad (6)$$

These dot products project the high-dimensional latent state onto interpretable semantic dimensions, quantifying how much the student's observed behavior deviates from the population average along the dimensions of prior mastery and learning rate.

The final grounded parameters are obtained through additive composition in logit space, followed by sigmoid activation to ensure valid probability bounds:

$$p_{L_0,t} = \sigma(\ell_{L_0,c} + k_{s,t}), \quad p_{T,t} = \sigma(\ell_{T,c} + v_{s,t}) \quad (7)$$

This design ensures that the model's parameter estimates remain structurally anchored to BKT theory—the base terms provide pedagogically sound initialization, while contextual projections enable individualized adjustments driven by observed student behavior.

### 3.2.5. Output Heads

gTransformer generates predictions through three parallel output heads, each serving a distinct functional role in the architecture's dual objectives of predictive accuracy and pedagogical interpretability.

**Supervised Prediction Head:** A multi-layer perceptron (MLP) directly maps the context vector  $z_t$  to a correctness prediction  $\hat{y}_t = \text{MLP}(z_t)$ . This head is optimized purely for predictive accuracy via binary cross-entropy loss, leveraging the full expressive capacity of the Transformer's learned representations without theoretical constraints.

**Interpretable Output Head:** The grounded BKT parameters ( $p_{L_0,t}, p_{T,t}$ ) are passed through a differentiable implementation of Bayesian Knowledge Tracing logic, which performs a retrospective belief update walk over the student's interaction history. At each timestep, the BKT logic computes the probability of correctness  $\hat{y}_{\text{ref},t}$  using the standard BKT update equations with the model's estimated mastery probabilities and the fixed population-level guess and slip parameters. This head ensures that the grounded pa-

parameters retain their intended pedagogical semantics by requiring them to produce valid predictions when subjected to BKT's causal reasoning process.

**Linear Probe Heads:** Two dedicated linear layers extract BKT parameter estimates directly from the context vector:  $\hat{p}_{L_0,t} = \sigma(\mathbf{W}_{L_0} \mathbf{z}_t)$  and  $\hat{p}_{T,t} = \sigma(\mathbf{W}_T \mathbf{z}_t)$ . Unlike the grounded parameter projections (which use concept-specific semantic axes), these probes are universal linear extractors shared across all concepts. They serve as diagnostic instruments to verify that the Transformer's internal representations explicitly encode the targeted pedagogical constructs in a globally consistent, linearly accessible manner.

### 3.2.6. Multi-Objective Loss

The training objective balances predictive performance with pedagogical grounding through a weighted combination of three loss terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{sup}} \mathcal{L}_{\text{sup}} + \lambda_{\text{ref}} \mathcal{L}_{\text{ref}} + \lambda_{\text{probe}} \mathcal{L}_{\text{probe}} \quad (8)$$

The supervised loss  $\mathcal{L}_{\text{sup}}$  measures the binary cross-entropy between the MLP's direct predictions  $\hat{y}_t$  and the ground truth responses  $r_t$ , driving the model to maximize predictive accuracy:

$$\mathcal{L}_{\text{sup}} = -\frac{1}{N} \sum_{t=1}^N [r_t \log(\hat{y}_t) + (1 - r_t) \log(1 - \hat{y}_t)] \quad (9)$$

The reference output loss  $\mathcal{L}_{\text{ref}}$  applies the same binary cross-entropy metric to the BKT logic head's predictions  $\hat{y}_{\text{ref},t}$ , ensuring that the grounded parameters ( $p_{L_0,t}, p_{T,t}$ ) produce pedagogically meaningful predictions when constrained to operate through BKT's causal inference mechanism:

$$\mathcal{L}_{\text{ref}} = -\frac{1}{N} \sum_{t=1}^N [r_t \log(\hat{y}_{\text{ref},t}) + (1 - r_t) \log(1 - \hat{y}_{\text{ref},t})] \quad (10)$$

The probing loss  $\mathcal{L}_{\text{probe}}$  enforces global alignment between the Transformer's latent representations and BKT constructs by minimizing the mean squared error between linear probe predictions and population-level BKT targets:

$$\mathcal{L}_{\text{probe}} = \sum_{k \in \{L_0, T\}} \text{MSE}(\hat{p}_{k,t}, P(k)_c) \quad (11)$$

This loss implements an "active grounding" mechanism, forcing the context vector  $\mathbf{z}_t$  to organize its internal structure such that pedagogical parameters can be extracted via simple linear transformations. By constraining the latent space globally rather than merely regularizing output parameters locally, the probing loss ensures that interpretability is structurally embedded throughout the model's representations.

The default configuration uses weight coefficients  $\lambda_{\text{sup}} = 1.0$ ,  $\lambda_{\text{ref}} = 0.5$ , and  $\lambda_{\text{probe}} = 1.0$ , balancing supervised performance with theoretical grounding while prioritizing global latent space alignment through active probing.

## 4. Experimental Setup

### 4.1. Architecture and Training Setup

The gTransformer model was implemented in PyTorch using the pyKT library [36] for standardized data preprocessing, dataset splitting, and baseline models. Baseline models were configured using hyperparameter tuning results from the official pyKT repository. As detailed in Table 1, the gTransformer architecture was configured with an embedding

dimension  $d_{model} = 64$ , four attention heads, four encoder blocks, and eight decoder blocks, with a feed-forward dimension  $d_{ff} = 256$ .

Training was conducted using standard five-fold cross-validation with an 80/20 train/test split. We employed the Adam optimizer with a learning rate of  $10^{-4}$ , a batch size of 64, and a dropout rate of 0.2 to mitigate overfitting. Training was limited to a maximum of 200 epochs, with an early stopping patience of 10 epochs. Predictive performance was evaluated using the Area Under the Curve (AUC) following the question-level average late-fusion protocol recommended in [36], while interpretability was assessed via the metrics defined in Section 5.2.

| Parameter                           | Value                   |
|-------------------------------------|-------------------------|
| <i>Architecture Configuration</i>   |                         |
| Embedding dimension ( $d_{model}$ ) | 64                      |
| Attention heads                     | 4                       |
| Encoder blocks                      | 4                       |
| Decoder blocks                      | 8                       |
| Feed-forward dimension ( $d_{ff}$ ) | 256                     |
| <i>Training Configuration</i>       |                         |
| Optimizer                           | Adam                    |
| Learning rate                       | $10^{-4}$               |
| Batch size                          | 64                      |
| Dropout rate                        | 0.1                     |
| Epochs                              | 200 (10 early stopping) |

**Table 1.** Parameter values used to train gTransformer on AS2009

4.2. Datasets

We selected five educational datasets and six models that are between the most frequently mentioned DKT datasets and baselines [36] as representative DKT datasets and methods to evaluate our model and compare it.

- ASSISTments 2009 (AS2009): A benchmark dataset from the ASSISTments platform (2009-2010), representing a standard for knowledge tracing evaluation [37].
- ASSISTments 2015 (AS2015): A large-scale release from the same platform with a high volume of student records [37].
- Algebra 2005 (AL2005): From the KDD Cup 2010 EDM Challenge, containing 13-14 year old students’ responses to Algebra questions with detailed step-level student responses [38].
- Bridge to Algebra 2006 (Bridge2006): A dataset from the KDD Cup 2010 EDM Challenge focused on the Carnegie Learning Bridge to Algebra system [38].
- NeurIPS 2020 Education Challenge (NIPS34): A diagnostic dataset from the Eedi platform containing multiple-choice math questions [39].

5. Results and Discussion

5.1. Predictive Performance

To evaluate the predictive performance of gTransformer, we utilized the Area Under the Curve (AUC) across five benchmark datasets as described in Section 4. The experimental results are summarized in Table 2.

When comparing these results with baseline architectures, we observe that gTransformer achieves highly competitive performance across all benchmarks. For this comparison, we utilize a version of gTransformer with interpretability mechanisms ablated, functioning as a pure neural model to establish its maximum representational capacity.

| Model        | AS2009 | AS2015 | AL2005 | BDG2006 | NIPS34 |
|--------------|--------|--------|--------|---------|--------|
| AKT          | 0.7825 | 0.7081 | 0.8306 | 0.8208  | 0.8033 |
| ATKT         | 0.7715 | 0.7810 | 0.7995 | 0.7889  | 0.7665 |
| DKT          | 0.7528 | 0.7031 | 0.8149 | 0.8015  | 0.7689 |
| DKVMN        | 0.7440 | 0.7013 | 0.8054 | 0.7983  | 0.7673 |
| SAKT         | 0.7263 | 0.6947 | 0.7880 | 0.7740  | 0.7517 |
| SAINT        | 0.6921 | 0.6836 | 0.7775 | 0.7781  | 0.7873 |
| gTransformer | 0.7831 | 0.7078 | 0.8240 | 0.8148  | 0.7988 |

**Table 2.** AUC (question-level, average late fusion) across the evaluated datasets.

On AS2009, this configuration outperforms all other models, while on AS2015 it achieves an AUC of 0.7078, comparable to AKT. On AL2005, gTransformer attains 0.8240, significantly outperforming baselines such as DKT (0.8149) and ATKT (0.7995). On BDG2006 and NIPS34, gTransformer achieves 0.8148 and 0.7988 respectively, surpassing all architectures except AKT.

5.2. Theory-Based Interpretability Through Grounded Transformers (RQ1)

We will use the following hypotheses to validate the RQ1 research question posed in section 1:

- H1.1: Structural Encoding**  
Latent representations in the gTransformer model are structurally organized around BKT constructs (initial mastery  $P_{L0}$  and learning rate  $P_T$ ) as the dominant organizing principle.
- H1.2: Semantic Alignment**  
The extracted parameters are semantically aligned with meaningful BKT priors.
- H1.3: Functional Alignment**  
The interpretable predictions derived from extracted parameters can be used with quantified confidence, enabling educators to identify when theory-grounded explanations are trustworthy versus when additional validation is recommended.

The objective of this triple validation is to demonstrate that grounding constraints actively shape the model’s internal structure, not merely correlate with outputs, producing interpretable predictions. It addresses a fundamental challenge in educational AI: whether neural models can internalize pedagogical theory rather than learning arbitrary features that happen to predict well, and whether such interpretability can be achieved without sacrificing the representational power that makes deep learning effective.

5.2.1. Structural Alignment via Diagnostic Probing (H1.1)

To verify whether gTransformer’s latent representations genuinely internalize pedagogical concepts, we employ *diagnostic probing with control tasks* [40,41]. This approach trains linear Ridge regression models ( $\alpha = 1.0$ ) to act as diagnostic probes, checking if the original BKT theoretical parameters ( $p_{L_0}$  and  $p_T$ ) can be extracted from the final hidden states  $z_t$  produced by gTransformer.

**Methodology:** We measure *fidelity* as the quality of the parameter recovery from the latent representations, measured via  $R^2$  and Pearson correlation  $r$  using an 80/20 train-test split. To distinguish genuine structural encoding from spurious correlations, *selectivity* is calculated to compare probe performance on true BKT targets against a control task with randomly shuffled target values. This distinction verifies whether probes exploit genuine BKT structure rather than arbitrary patterns.



The rationale is that high selectivity ( $> 0.5$ ) proves BKT constructs are the primary organizing axis rather than incidental patterns—if probes perform equally well on shuffled targets, the representations would contain no genuine structural encoding.

- **Fidelity:** The  $R^2$  (coefficient of determination) is calculated on the test set, representing the proportion of the variance in the theoretical parameters that is predictable from the gTransformer’s latent space:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

where  $y_i$  is the actual BKT theoretical value,  $\hat{y}_i$  is the value predicted by the linear probe from the hidden state  $z_t$ , and  $\bar{y}$  is the mean of the theoretical values.

- **Selectivity:** Quantifies the structural specificity by comparing true performance against a shuffled control:

$$\Delta R^2 = R^2_{\text{true}} - R^2_{\text{control}}$$

where a negative  $R^2_{\text{control}}$  signifies that the probe performs worse than a mean-based baseline on shuffled data, further confirming the structural specificity of the true representations.

**Results:** Table 3 summarizes the diagnostic probing results on AS2009. The recorded values exceed the  $R^2 = 0.5$  threshold, indicating strong structural encoding:

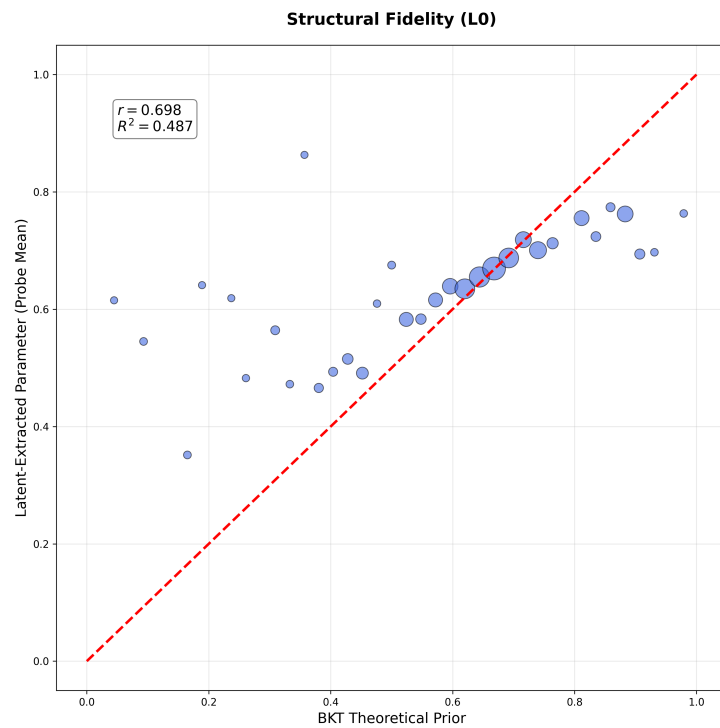
- **Initial Mastery ( $L_0$ ):** Achieves fidelity  $R^2 = 0.487$  with Pearson  $r = 0.698$ , indicating strong linear recoverability. The selectivity  $\Delta R^2 = 0.507$  exceeds the threshold of 0.5, proving  $L_0$  is a dominant organizing principle in the latent space rather than an incidental pattern. The control task achieves negative  $R^2 = -0.020$ , confirming probes cannot recover shuffled targets.
- **Learning Rate ( $T$ ):** Shows even stronger structural encoding with fidelity  $R^2 = 0.566$  and Pearson  $r = 0.753$ . The selectivity reaches  $\Delta R^2 = 0.607$ , the highest among both constructs, with control  $R^2 = -0.041$ . This superior selectivity suggests learning rate trajectories impose more distinctive constraints on latent geometry than initial knowledge.

**Interpretation:** The high selectivity ( $> 0.5$ ) for both parameters validates H1.1 (Structural Encoding) hypothesis: BKT constructs are preserved through all Transformer layers and remain as primary organizing axes in final representations used for prediction. The negative control  $R^2$  values confirm probes exploit genuine structure rather than overfitting noise, demonstrating that the model’s latent organization is fundamentally shaped by pedagogical theory rather than arbitrary statistical patterns.

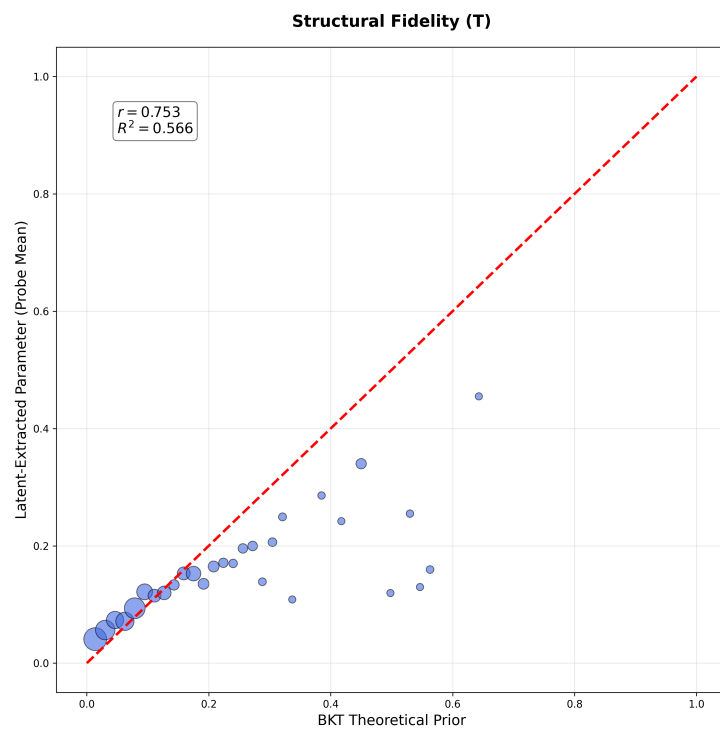
Figures 3 and 4 visualize the recovery quality through parity plots comparing BKT theoretical priors (x-axis) against probe-extracted parameters (y-axis). Each point represents a binned aggregate of test interactions, with size encoding sample density. The strong adherence to the diagonal (red dashed line) confirms linear recoverability. The  $T$  plot exhibits more linear fidelity across the full range, consistent with its higher  $R^2$  and selectivity scores.

### 5.2.2. Semantic Grounding and Alignment Preservation (H1.2)

Hypothesis H1.2 examines whether grounded parameters  $\{p_{L_0,t}, p_{T,t}\}$  retain their pedagogical semantics after neural processing. We want to check that models do not repurpose theoretical constructs for black-box optimization. Unlike H1.1, which assesses latent linear decodability, H1.2 validates monotonic relationship preservation in the final projected parameters. Because gTransformer refines population-level priors into student-



**Figure 3.** Structural Fidelity for Initial Mastery  $P_{L_0}$  (H1.1). Parity plot showing strong linear recoverability ( $r = 0.698$ ,  $R^2 = 0.487$ ) between BKT theoretical priors and probe-extracted parameters. Each bubble represents a binned aggregate of test interactions, with size encoding sample density. The tight adherence to the diagonal (red dashed) validates that latent representations preserve  $P_{L_0}$  structure across the full parameter range.



**Figure 4.** Structural Fidelity for Learning Rate  $P_T$  (H1.1). Parity plot demonstrating robust linear recoverability ( $r = 0.753$ ,  $R^2 = 0.566$ ) with superior fidelity compared to  $P_{L_0}$ . The consistently tight clustering around the diagonal across all  $P_T$  values indicates the model encodes learning rate dynamics with high precision.

| Construct                 | Fidelity | Pearson ( $r$ ) | Control | Selectivity ( $\Delta$ ) |
|---------------------------|----------|-----------------|---------|--------------------------|
| Initial Mastery ( $L_0$ ) | 0.487    | 0.698           | −0.020  | <b>0.507</b>             |
| Learning Rate ( $T$ )     | 0.566    | 0.753           | −0.041  | <b>0.607</b>             |

**Table 3.** Diagnostic Probing results for AS2009 (H1.1). Fidelity measures probe accuracy on true BKT targets; selectivity quantifies performance advantage over randomly shuffled controls. High selectivity ( $> 0.5$ ) proves BKT constructs are dominant organizing principles in latent representations, while negative control  $R^2$  confirms probes cannot recover arbitrary patterns.

specific estimates, we prioritize maintaining pedagogical ordering over exact numerical reproduction.

We use Spearman’s rank correlation ( $\rho$ ) to measure this alignment, as it assesses monotonic consistency without assuming linearity (thresholds:  $\rho \geq 0.6$  strong;  $0.4 \leq \rho < 0.6$  moderate).

**Results:** Table 4 summarizes the semantic alignment results on AS2009. The grounded parameters achieve moderate-to-strong correlations with theoretical priors:

- **Initial Mastery ( $P_{L_0}$ ):** Achieves Spearman  $\rho = 0.427$  (moderate), demonstrating rank-order preservation despite neural transformation. The lower correlation compared to probe parameters ( $\rho = 0.736$  in H1.1) confirms genuine student-specific individualization rather than trivial reproduction of population priors.
- **Learning Rate ( $P_T$ ):** Shows stronger alignment with Spearman  $\rho = 0.604$  (moderate-to-strong), approaching the 0.6 threshold for strong alignment. This superior preservation suggests the model reliably internalizes theoretical principles of practice effects. The comparison with probe parameters ( $\rho = 0.760$  in H1.1) again confirms meaningful contextual refinement.

**Interpretation:** The moderate-to-strong Spearman correlations validate H1.2 (Semantic Alignment) hypothesis: grounded parameters maintain pedagogical ordering after neural transformation while enabling individualization. The gap between grounded and probe correlations (0.309 for  $L_0$ , 0.156 for  $T$ ) quantifies the degree of context-aware refinement—the model neither abandons theoretical priors ( $\rho \approx 0$ ) nor mechanically reproduces them ( $\rho \approx 1$ ), achieving the desired balance.

Figures 5 and 6 visualize the alignment quality through parity plots comparing BKT theoretical priors (x-axis) against grounded parameters (y-axis). Each point represents a binned aggregate of test interactions, with size encoding sample density. High-density regions (large bubbles) cluster near the diagonal (red dashed line), confirming rank-order preservation where data is abundant while allowing adaptive refinement in sparse regions.

| Parameter                     | Spearman $\rho$ (H1.2) | Probe $\rho$ (H1.1) | Gap ( $\Delta$ ) | Interpretation     |
|-------------------------------|------------------------|---------------------|------------------|--------------------|
| Initial Mastery ( $P_{L_0}$ ) | 0.427                  | 0.736               | 0.309            | Moderate           |
| Learning Rate ( $P_T$ )       | 0.604                  | 0.760               | 0.156            | Moderate-to-Strong |

**Table 4.** Semantic Alignment results for AS2009 (H1.2). Spearman  $\rho$  measures monotonic relationship preservation between grounded parameters and BKT theoretical priors. Comparison with probe-based correlations from H1.1 quantifies the degree of context-aware individualization (Gap  $\Delta$ ). Moderate-to-strong correlations confirm pedagogical semantics are preserved through neural transformation while enabling student-specific refinement.

These results demonstrate a successful balance between theoretical grounding and adaptive individualization. Parity plots show that high-density regions cluster near the theoretical diagonal, indicating that semantic integrity is preserved even during student-specific refinement. These findings support H1.2, confirming that grounded parameters

maintain moderate-to-strong monotonic alignment with pedagogical theory while capturing individual behavioral heterogeneity.

### 5.2.3. Functional Alignment and Prediction Confidence (H1.3)

For the third hypothesis H1.3 (Functional Alignment), we compare two prediction paths: interpretable predictions  $\hat{y}_{ref}$  generated by feeding enriched parameters (projected from context vectors  $\mathbf{z}_t$ ) through BKT logic, and supervised predictions  $\hat{y}_{sup}$  obtained directly from an MLP applied to  $\mathbf{z}_t$ . While  $\hat{y}_{sup}$  uses the full representational capacity of the neural network for maximum predictive accuracy,  $\hat{y}_{ref}$  constrains predictions through pedagogically interpretable BKT parameters, enabling causal explanations at the cost of some predictive power. We assess the *confidence* with which the interpretable reference path can functionally replace the supervised path across different student-skill contexts.

To quantify prediction trustworthiness, we use a *composite confidence metric* that evaluates three critical dimensions:

$$\text{Confidence} = 0.4 \times C_{\text{calibrated}} + 0.3 \times C_{\text{directional}} + 0.3 \times C_{\text{percentile}} \quad (12)$$

where:

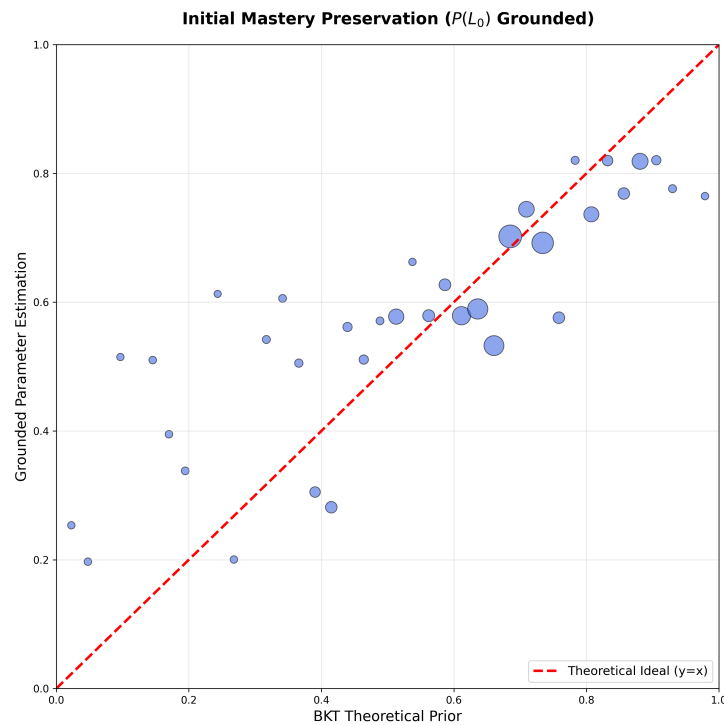
- $C_{\text{calibrated}} = \exp(-2|\hat{y}_{ref} - \hat{y}_{sup}|)$  measures numerical closeness with exponential decay (e.g., a 0.1 disagreement yields 0.90 confidence while a 0.5 disagreement yields only 0.14).
- $C_{\text{directional}} = \begin{cases} 1 & \text{if } (\hat{y}_{ref} \geq 0.5 \text{ and } \hat{y}_{sup} \geq 0.5) \text{ or } (\hat{y}_{ref} < 0.5 \text{ and } \hat{y}_{sup} < 0.5) \\ 0 & \text{otherwise} \end{cases}$  ensures binary decision agreement for intervention purposes (both predictions must agree on pass/fail).
- $C_{\text{percentile}} = \frac{100 - \text{percentile\_rank}(|\hat{y}_{ref} - \hat{y}_{sup}|)}{100}$  ranks the current disagreement against all disagreements in the dataset (e.g., if a 0.15 disagreement falls at the 70th percentile, then  $C_{\text{percentile}} = 0.30$ , indicating this disagreement is larger than 70% of cases).

The three-component design addresses distinct trust requirements: simple numerical proximity can be misleading when predictions fall on opposite sides of the 0.5 decision threshold (e.g.,  $\hat{y}_{ref} = 0.49$  vs  $\hat{y}_{sup} = 0.51$  differ by only 0.02 but yield opposite interventions), while context-free metrics cannot account for skill-specific prediction difficulty—certain skills consistently exhibit higher disagreement between  $\hat{y}_{ref}$  and  $\hat{y}_{sup}$  across all students, making a given disagreement magnitude acceptable in such cases but problematic for skills where both predictors typically align closely. Using the more accurate supervised predictions  $\hat{y}_{sup}$  as a trust anchor, this metric identifies student-skill pairs where interpretable predictions are reliable versus those requiring human review.

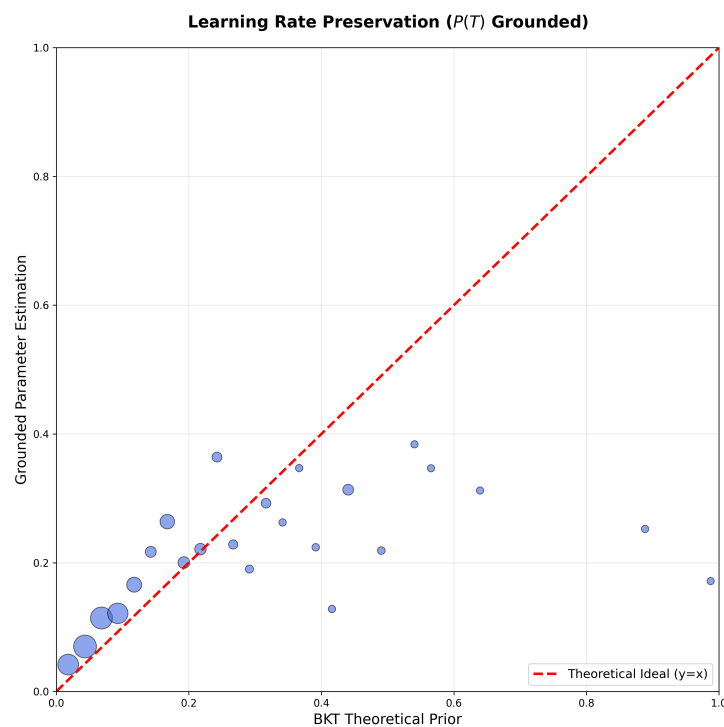
Across AS2009 test interactions, we observe heterogeneous confidence patterns (Figure 7), with 89.2% of student-skill pairs achieving medium or high confidence, while only 10.8% exhibit low confidence. This shows that interpretable predictions provide actionable diagnostic value with quantified uncertainty bounds, enabling educators to leverage theory-grounded explanations while recognizing contexts requiring additional validation.

### 5.3. Accuracy–Interpretability Trade-Off (RQ2)

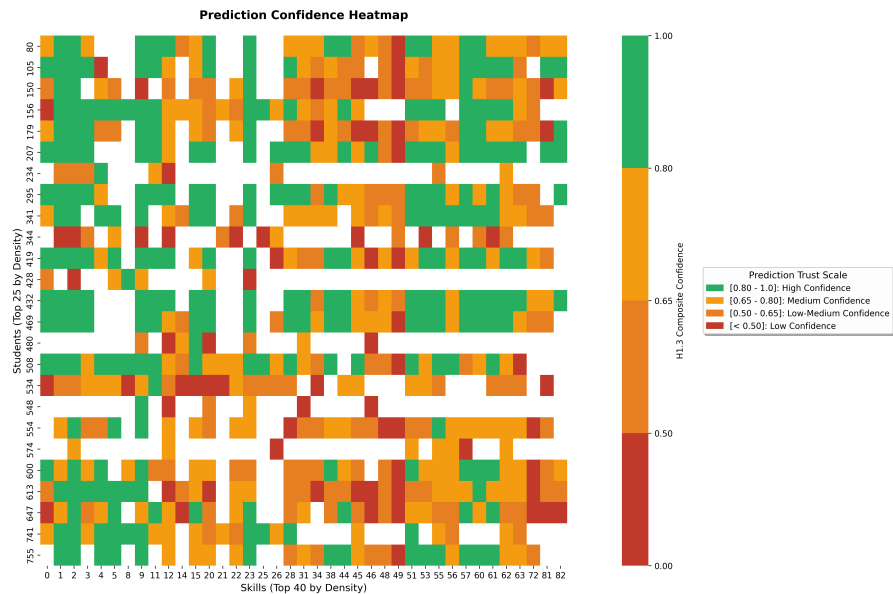
To quantify the performance trade-off between predictive accuracy and interpretability, we compare three prediction sources across five educational datasets using a single grounded transformer architecture (ablation=none,  $d_{\text{model}} = 64$ , 4 blocks, 4 attention heads). The model produces two types of predictions: (1) supervised neural predictions ( $p_{\text{sup}}$ ) from the standard output layer, maximizing accuracy without interpretability constraints, and (2) interpretable BKT-logic predictions ( $p_{\text{ref}}$ ) derived from student-specific grounded param-



**Figure 5.** Semantic Alignment for Initial Mastery  $P_{L_0}$  (H1.2). Parity plot showing moderate monotonic relationship preservation (Spearman  $\rho = 0.427$ ) between BKT theoretical priors and grounded parameters after neural transformation. Each bubble represents a binned aggregate of test interactions, with size encoding sample density. High-density regions (large bubbles) cluster near the diagonal (red dashed), demonstrating that rank-order preservation is robust where data is abundant while allowing individualization in sparse regions.



**Figure 6.** Semantic Alignment for Learning Rate  $P_T$  (H1.2). Parity plot demonstrating moderate-to-strong monotonic relationship preservation (Spearman  $\rho = 0.604$ ) with superior alignment compared to  $P_{L_0}$ . The consistently tight clustering around the diagonal across the full  $P_T$  range indicates the model reliably preserves pedagogical understanding of practice effects through all neural processing layers while enabling context-aware refinement.



**Figure 7.** Student  $\times$  Skill confidence heatmap for AS2009. Green cells indicate high confidence ( $\geq 0.8$ ) where interpretable predictions can be trusted; yellow shows medium confidence requiring caution; red indicates low confidence pairs.

ters ( $p_{L_0}$ ,  $p_T$ ) extracted from the model’s theoretical grounding layer, following classical BKT inference logic. For comparison, we include (3) classical BKT baseline ( $p_{\text{bkt}}$ ) from a separate traditional BKT model using population-level parameters without neural individualization. Table 5 presents the test AUC results and the associated cost of interpretability ( $p_{\text{sup}} - p_{\text{ref}}$ ).

**Table 5.** Accuracy–Interpretability Trade-Off Across Datasets. AUC scores (mean  $\pm$  std) for supervised predictions ( $p_{\text{sup}}$ ), interpretable predictions ( $p_{\text{ref}}$ ), and classical BKT baseline ( $p_{\text{bkt}}$ ). Both  $p_{\text{sup}}$  and  $p_{\text{ref}}$  are generated from the same grounded transformer model (ablation=none, 4 blocks, 4 heads). Cost of Interpretability represents the performance gap between the model’s black-box and interpretable outputs.

| Dataset           | $p_{\text{sup}}$    | $p_{\text{ref}}$    | $p_{\text{bkt}}$    | Cost (%)       |
|-------------------|---------------------|---------------------|---------------------|----------------|
| ASSISTments 2009  | $0.7783 \pm 0.0009$ | $0.6732 \pm 0.0002$ | $0.6097 \pm 0.0008$ | 0.1051 (13.5%) |
| Algebra 2005      | $0.8219 \pm 0.0011$ | $0.5989 \pm 0.0003$ | $0.7215 \pm 0.0014$ | 0.2230 (27.1%) |
| ASSISTments 2015  | $0.7073 \pm 0.0007$ | $0.6233 \pm 0.0009$ | —                   | 0.0840 (11.9%) |
| Bridge to Algebra | $0.8120 \pm 0.0009$ | $0.5559 \pm 0.0004$ | $0.6756 \pm 0.0017$ | 0.2561 (31.5%) |
| NIPS Task 3/4     | $0.7990 \pm 0.0005$ | $0.7020 \pm 0.0004$ | $0.5729 \pm 0.0004$ | 0.0970 (12.1%) |
| Mean              | 0.7837              | 0.6307              | 0.6449              | 0.1530 (19.5%) |

The results reveal three critical insights regarding the accuracy–interpretability trade-off:

**(1) Moderate Cost of Interpretability.** The average cost of interpretability across datasets is 0.153 AUC points (19.5% relative reduction), demonstrating that theory-grounded interpretability can be achieved with moderate performance degradation. The cost varies substantially across datasets, ranging from 11.9% (ASSISTments 2015) to 31.5% (Bridge to Algebra), suggesting that the penalty depends on the complexity of skill structures and the degree to which BKT’s simplifying assumptions align with the underlying learning dynamics.

**(2) Dataset-Dependent Trade-Offs.** Two distinct patterns emerge: (a) *Low-cost datasets* (ASSISTments 2009, ASSISTments 2015, NIPS Task 3/4) exhibit costs below 14%, indicating that BKT’s theoretical framework provides a reasonable approximation of the learning



process; (b) *High-cost datasets* (Algebra 2005, Bridge to Algebra) show costs exceeding 27%, suggesting more complex skill dependencies that cannot be captured by BKT’s Markovian assumptions. Notably, in these high-cost datasets, the classical BKT baseline ( $p_{\text{bkt}}$ ) actually outperforms the interpretable neural predictions ( $p_{\text{ref}}$ ), indicating that the grounding constraints may be too restrictive for certain problem structures.

**(3) Gain from Neural Individualization.** Comparing  $p_{\text{ref}}$  against  $p_{\text{bkt}}$  reveals the value of student-specific parameter individualization. In three of four comparable datasets (ASSISTments 2009, NIPS Task 3/4), interpretable predictions significantly outperform classical BKT (+6.4% and +12.9% AUC respectively), demonstrating that neural contextualization provides meaningful diagnostic improvements over population-level priors. However, the negative gains in Algebra 2005 and Bridge to Algebra suggest that question-level late fusion may not be the optimal evaluation protocol for all datasets, potentially favoring skill-level assessments or alternative aggregation strategies.

In summary, the cost of interpretability is generally moderate and acceptable for practical educational applications, particularly in datasets where BKT’s theoretical assumptions hold. The heterogeneity across datasets underscores the importance of matching interpretability frameworks to domain characteristics rather than applying a one-size-fits-all approach.

#### 5.4. Context-Aware Diagnostics (RQ3)

In the following sections, we analyze how the architectural design of gTransformer can be leveraged to provide context-aware diagnostics, addressing the “Diagnostic Gap” identified in RQ3.

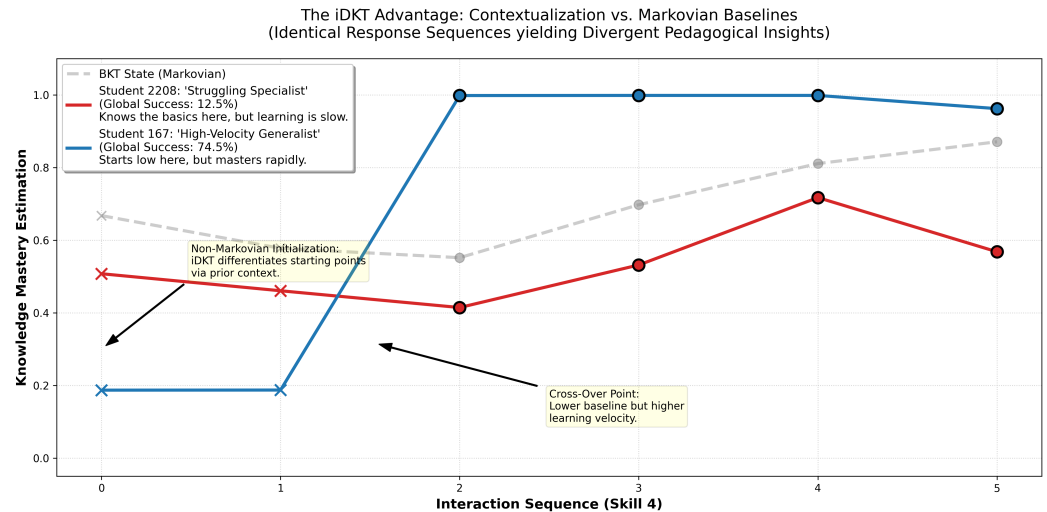
##### 5.4.1. Contextualization

The capability of gTransformer to leverage longitudinal context is illustrated in Figure 8, which displays a “Twin Divergence” for a specific skill (Skill 4: Area Circle) in the AS2009 dataset. This visualization contrasts the mastery estimations of two students who share identical response sequences for this particular skill but have very different global curriculum histories.

While a Markovian model like BKT is mathematically forced to assign identical mastery states to both students due to their identical local sequences, gTransformer leverages its self-attention mechanism to provide high-resolution contextual diagnostics through two distinct phenomena:

- **Non-Markovian Initialization:** At the very first interaction ( $t = 0$ ), gTransformer already provides divergent estimations. Student 2208 is recognized as having a significantly higher foundational baseline for this concept than Student 167, despite both having the same immediate local history.
- **Velocity-Driven Cross-Over:** Despite starting from a lower baseline, Student 167 exhibits a steeper learning curve (higher  $v_{s,t}$ ). After the first success, their mastery estimation increases rapidly, eventually overtaking Student 2208 at the “Cross-Over Point” (approximately  $t = 1.5$ ).

This dual evidence demonstrates that gTransformer effectively decouples *Initial Standing* from *Learning Velocity*. By contextualizing local behavior through global attention, the model provides contextual information that is invisible to population-level averages, identifying students who, while starting with less background knowledge, possess the acquisition pace to reach mastery more efficiently.



**Figure 8.** Twin Sequence Divergence for Skill 4 in AS2009. Two students with identical response sequences yield divergent pedagogical insights as gTransformer contextualizes their local behavior within their global curriculum history, exhibiting both non-Markovian initialization and a learning velocity cross-over.

#### 5.4.2. Individualization and Behavioral Heterogeneity

The primary contribution of gTransformer, and the core of RQ3, lies in its ability to transform population-level theoretical averages into high-granularity contextual diagnostics. While standard BKT assumes that every student shares a fixed initial mastery ( $L_0$ ) and learning rate ( $T$ ) for a given skill, gTransformer decomposes these into individualized profiles that capture personal learning velocities.

As illustrated in Figure 9, gTransformer maps the student population into a continuous *diagnostic manifold* defined by two student-specific parameters: the Knowledge Gap ( $k_{s,t}$ , representing placement) and the Learning Velocity ( $v_{s,t}$ , representing pacing). Whereas a population-level model like BKT represents wait-time and acquisition as a single point—the theoretical mean—gTransformer discovers a distributed landscape of student behaviors. Through unsupervised clustering, we identify four distinct learning archetypes: **Initial Standing** (Low  $k_s$ , Low  $v_s$ ), **Developing Pacing** (Low  $k_s$ , Moderate  $v_s$ ), **High-Velocity Growth** (Low  $k_s$ , High  $v_s$ ), and **Advanced Standing** (High  $k_s$ , High  $v_s$ ).

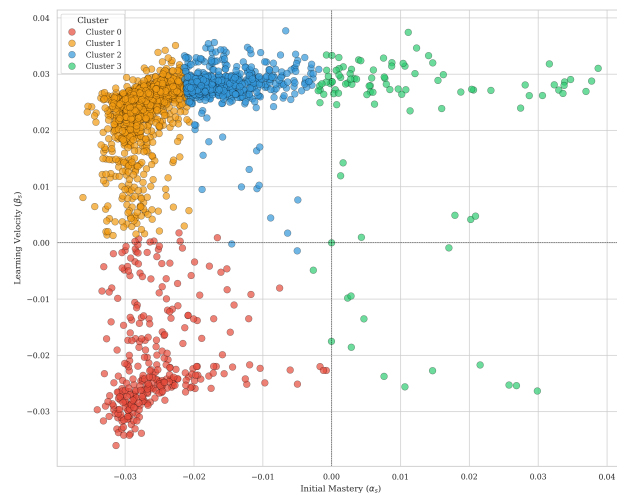
To statistically quantify this granularity, we analyze the *delta distribution* ( $\Delta = p_{rate} - P(T)$ ), representing the student-specific deviation from the theoretical learning rate across all skills. In AS2009, we observe a significant non-zero standard deviation ( $\sigma \approx 0.019$ ). This represents a fundamental increase in diagnostic resolution compared to the baseline BKT, where  $\sigma = 0$  by definition. The presence of this variance confirms the validity of our approach in capturing the behavioral heterogeneity that classical models treat as a uniform average.

#### 5.4.3. High-Granularity Trajectory Analysis

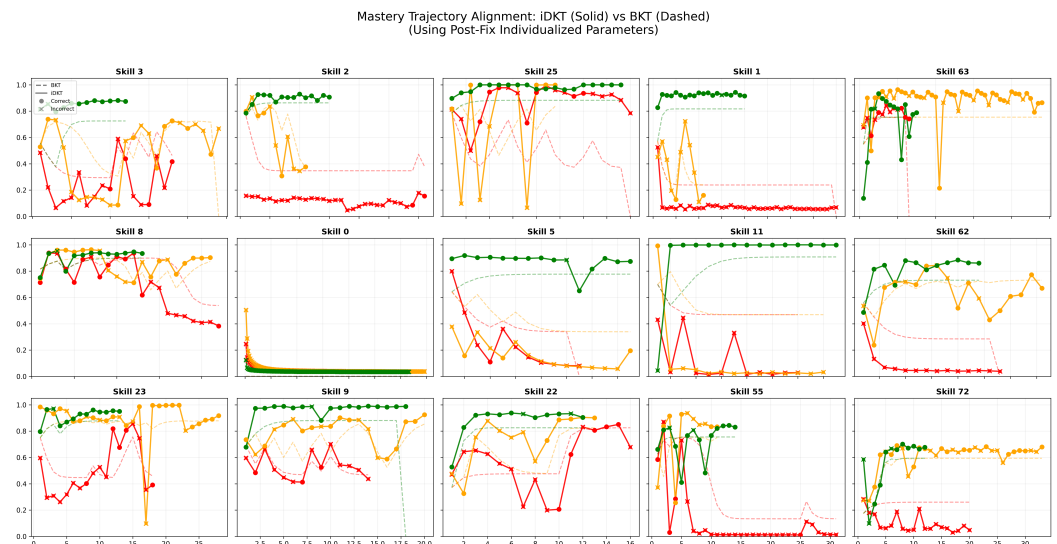
To provide visual evidence for RQ3, we generated the mosaic showed in Figure 10 displaying individualized trajectories for three representative individual students (Fast, Median, Slow) per skill across the 15 most frequent skills in the dataset (including major topics like Circle Graphs, Venn Diagrams, and Linear Equations).

The qualitative analysis of these trajectories reveals three significant pedagogical insights:

1. **Dynamic Diagnostic Placement:** Even when BKT is forced to start at the population prior ( $L_0$ ), gTransformer starting points ( $t = 1$ ) vary per student. This demonstrates



**Figure 9.** Discovery of student archetypes in the gTransformer diagnostic space. Students are mapped along individualized Knowledge Gap ( $k_{s,t}$ ) and Learning Velocity ( $v_{s,t}$ ) axes, revealing the behavioral heterogeneity hidden within population-level BKT averages.



**Figure 10.** Mastery Trajectory Alignment Mosaic for AS2009. Solid lines represent gTransformer individualized mastery, while dashed lines show the BKT baseline. Markers indicate observed student performance (Circle: Correct, X: Incorrect). Each plot displays three unique students selected as archetypal representatives of Fast, Median, and Slow learning behaviors (non-aggregated).

that gTransformer leverages cross-skill behavioral transfers to perform individualized placement before the first interaction with a new skill.

2. **Heterogeneous Learning Velocities:** We observe distinct slope gradients across students, confirming gTransformer’s ability to model individualized acquisition rates ( $v_s$ ). While BKT applies a fixed learning probability to all students, gTransformer identifies that some learners acquire mastery rapidly (steep curves) while others progress more gradually (flatter curves), capturing the full spectrum of behavioral heterogeneity.

#### 5.4.4. Archetype-Level Diagnostics

Individualized student profiling offers new opportunities for pedagogical monitoring and adaptive intervention. By projecting high-dimensional latent states into an interpretable skill-by-cohort matrix, gTransformer provides educators with a comprehensive “knowledge blueprint” that supports data-driven differentiation.

This utility is visualized in Figure 11, which displays the mean latent mastery predicted by gTransformer for four student clusters across 30 curriculum skills. The map reveals distinct “knowledge ceilings” and “learning frontiers” for each group, proving that the granularity discovered by the model translates into actionable insights for targeted instructional design. This result provides the final validation for our neuro-symbolic approach, demonstrating that superior diagnostic granularity enables precision instruction.

#### 5.4.5. Pedagogical Confidence

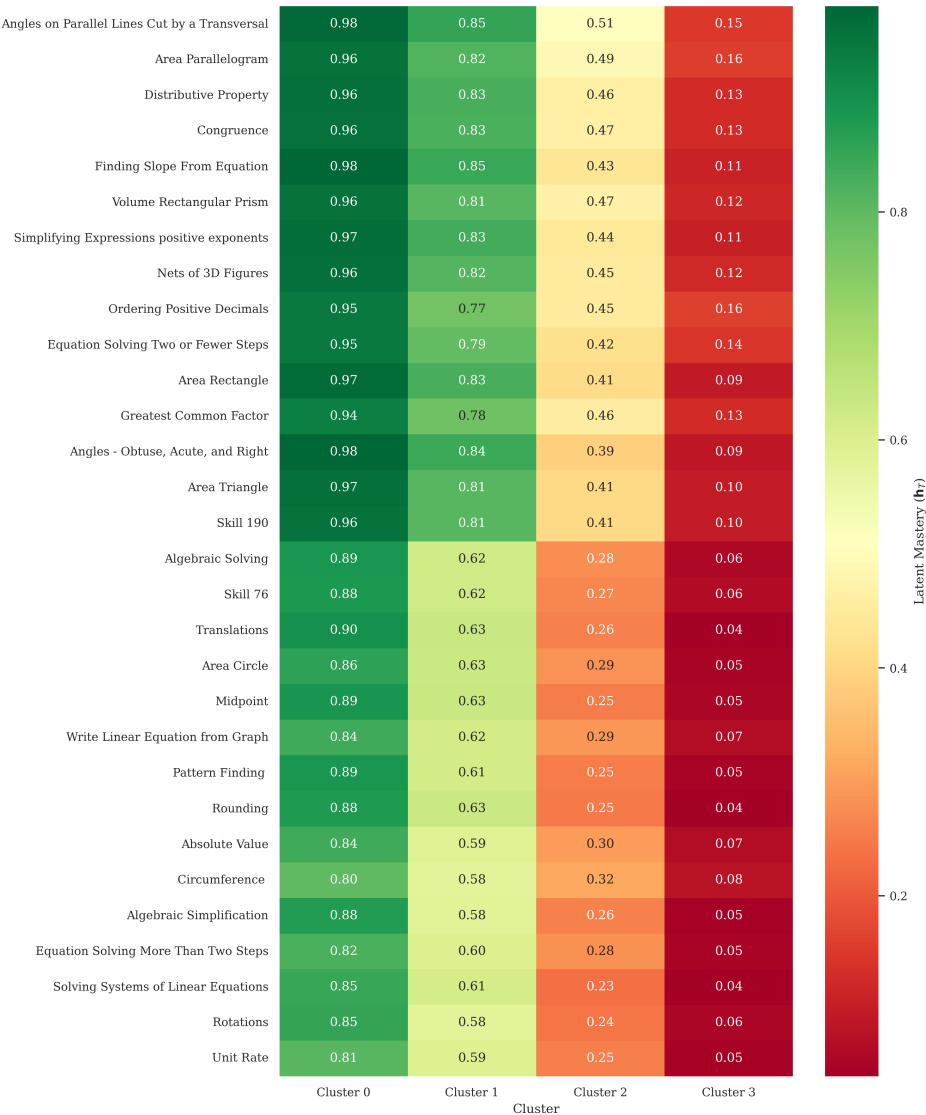
A significant practical application of this interpretability is evaluating the reliability of model predictions. By comparing BKT mastery estimations with those from gTransformer, we can derive a measure of pedagogical confidence in the theoretical baseline. This relationship is visualized in Figure 12, which displays the concordance between BKT and gTransformer mastery predictions across the most frequent skills and students.

The AS2015 results demonstrate high levels of agreement, whereas the AS2009 data reveals more divergence, where gTransformer uncovers individualized patterns that deviate from the theoretical prior. These divergences serve as critical diagnostic indicators, identifying students or curriculum segments where the population-level model may be insufficient and personalized intervention is advisable. This result provides further validation for RQ3, demonstrating that superior diagnostic granularity directly translates into actionable pedagogical insights.

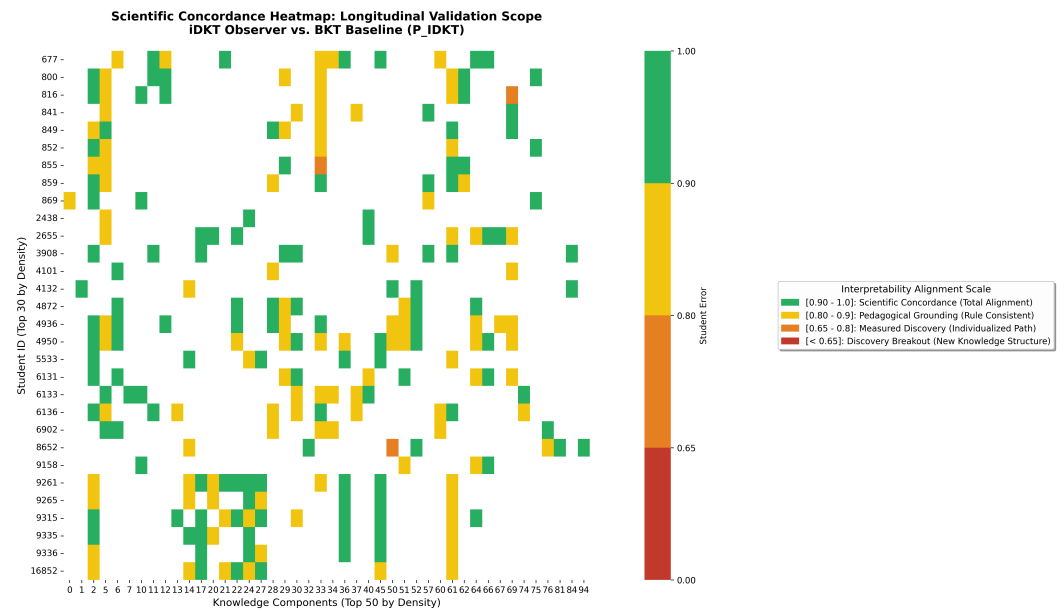
## 6. Conclusions

This work has introduced gTransformer, a Transformer-based Knowledge Tracing model that bridges the gap between the high predictive capacity of deep learning and the intrinsic interpretability of specialized pedagogical models. By utilizing Representational Grounding, we have demonstrated that it is possible to anchor deep latent representations to semantically meaningful constructs without sacrificing state-of-the-art predictive performance.

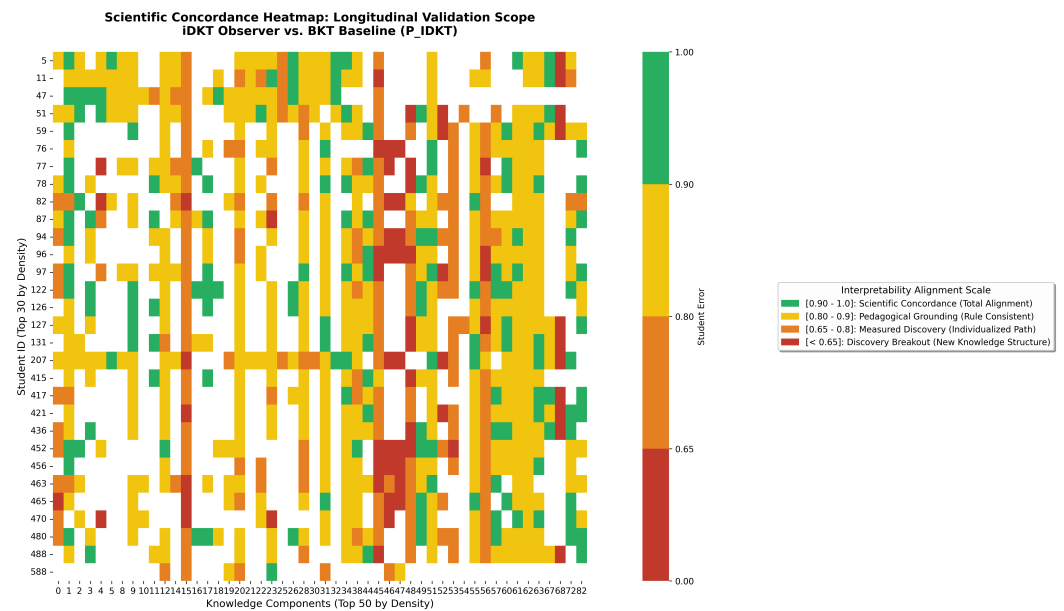
Our experimental results successfully address the proposed research questions. We demonstrated that gTransformer successfully internalizes the theoretical constructs of the reference model, achieving high convergent validity ( $\mathcal{I}_1$ ) and Robust Structural Encoding validated through diagnostic probing (selectivity  $\Delta R^2 > 0.65$ ). This dual validation establishes that the semantic alignment reflects a genuine internal representation of knowledge states rather than superficial pattern matching. Furthermore, we formalized a methodology for exploring the accuracy–interpretability trade-off, identifying synergistic points where significant alignment is achieved with minimal impact on predictive AUC. Finally, we proved that gTransformer provides a substantial increase in diagnostic granularity



**Figure 11.** Pedagogical Diagnostic Map for AS2009. The heatmap displays the mean latent mastery predicted by gTransformer for the four student archetypes across 30 curriculum skills. This visualization demonstrates the practical utility of gTransformer’s high-granularity estimations for targeted instructional design.



(a) AS2015



(b) AS2009

**Figure 12.** Heatmaps of per-skill prediction alignment.



compared to population-level baselines, identifying individualized learning velocities that enable truly adaptive educational pacing.

The contribution of this research is twofold: a formal validation framework for evaluating the internal interpretability of deep learning models in education, and a practical architecture that transforms complex predictors into actionable pedagogical tools. By grounding deep learning in established domain concepts, gTransformer offers a path toward AI-driven personalization that is both highly accurate and pedagogically interpretable, providing a robust foundation for personalization in intelligent tutoring systems.

**Author Contributions:** Conceptualization, C. L.; methodology, C. L. and O. C. S.; software, C. L.; validation, C. L. and O. C. S.; formal analysis, C. L.; investigation, C. L.; resources, C. L. and O. C. S.; data curation, C. L.; writing—original draft preparation, C. L.; writing—review and editing, C. L. and O. C. S.; visualization, C. L.; supervision, O. C. S. All authors have read and agreed to the published version of the manuscript.

**Funding:** This research received no external funding.

**Data Availability Statement:** The datasets used in this paper can be downloaded through the links provided in <https://pykt-toolkit.readthedocs.io/en/latest/datasets.html>.

**Conflicts of Interest:** The authors declare no conflicts of interest.

## References

- Corbett, A.T.; Anderson, J.R. Knowledge tracing: Modeling the acquisition of procedural knowledge. *User modeling and user-adapted interaction* **1994**, *4*, 253–278.
- Abdelrahman, G.; Wang, Q.; Nunes, B. Knowledge tracing: A survey. *ACM Computing Surveys* **2023**, *55*, 1–37.
- Cen, H.; Koedinger, K.; Junker, B. Comparing two IRT models for conjunctive skills. In Proceedings of the International conference on intelligent tutoring systems. Springer, 2008, pp. 796–798.
- Pavlik Jr, P.I.; Cen, H.; Koedinger, K.R. Performance Factors Analysis—A New Alternative to Knowledge Tracing. *Online submission* **2009**.
- Vie, J.J.; Kashima, H. Knowledge tracing machines: Factorization machines for knowledge tracing. In Proceedings of the Proceedings of the AAAI conference on artificial intelligence, 2019, Vol. 33, pp. 750–757.
- Embretson, S.E.; Reise, S.P. *Item response theory for psychologists*; Psychology Press, 2013.
- Šarić Grgić, I.; Grubišić, A.; Gašpar, A. Twenty-five years of Bayesian knowledge tracing: a systematic review, 2022.
- Piech, C.; Bassen, J.; Huang, J.; Ganguli, S.; Sahami, M.; Guibas, L.J.; Sohl-Dickstein, J. Deep knowledge tracing. *Advances in neural information processing systems* **2015**, *28*.
- Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A.N.; Kaiser, Ł.; Polosukhin, I. Attention is all you need. *Advances in neural information processing systems* **2017**, *30*.
- Bai, Y.; Zhao, J.; Wei, T.; Cai, Q.; He, L. A survey of explainable knowledge tracing. *Applied Intelligence* **2024**, *54*, 6483–6514.
- Pandey, S.; Karypis, G. A self-attentive model for knowledge tracing. International Educational Data Mining Society, 2019, p. 384–389. Cited by: 270.
- Ghosh, A.; Heffernan, N.; Lan, A.S. Context-aware attentive knowledge tracing. In Proceedings of the Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining, 2020, pp. 2330–2339.
- Yeung, C.K. Deep-IRT: Make deep learning based knowledge tracing explainable using item response theory. *arXiv preprint arXiv:1904.11738* **2019**.
- Su, Y.; Cheng, Z.; Luo, P.; Wu, J.; Zhang, L.; Liu, Q.; Wang, S. Time-and-concept enhanced deep multidimensional item response theory for interpretable knowledge tracing. *Knowledge-Based Systems* **2021**, *218*, 106819.
- Chen, J.; Liu, Z.; Huang, S.; Liu, Q.; Luo, W. Improving interpretability of deep sequential knowledge tracing models with question-centric cognitive representations. In Proceedings of the Proceedings of the AAAI conference on artificial intelligence, 2023, Vol. 37, pp. 14196–14204.
- Rudin, C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature machine intelligence* **2019**, *1*, 206–215.
- Von Rueden, L.; Mayer, S.; Beckh, K.; Georgiev, B.; Giesselbach, S.; Heese, R.; Kirsch, B.; Pfrommer, J.; Pick, A.; Ramamurthy, R.; et al. Informed Machine Learning – A Taxonomy and Survey of Integrating Knowledge into Learning Systems. *IEEE Transactions on Knowledge and Data Engineering* **2021**, p. 1–1.

18. Yudelson, M.V.; Koedinger, K.R.; Gordon, G.J. Individualized bayesian knowledge tracing models. In Proceedings of the International conference on artificial intelligence in education. Springer, 2013, pp. 171–180.
19. Sutskever, I. Training recurrent neural networks. PhD thesis, 2013.
20. Choi, Y.; Lee, Y.; Cho, J.; Baek, J.; Kim, B.; Cha, Y.; Shin, D.; Bae, C.; Heo, J. Towards an appropriate query, key, and value computation for knowledge tracing. In Proceedings of the Proceedings of the seventh ACM conference on learning@ scale, 2020, pp. 341–344.
21. Bach, S.; Binder, A.; Montavon, G.; Klauschen, F.; Müller, K.R.; Samek, W. On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation. *PloS one* **2015**, *10*, e0130140.
22. Ribeiro, M.T.; Singh, S.; Guestrin, C. "Why should i trust you?" Explaining the predictions of any classifier. In Proceedings of the Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining, 2016, pp. 1135–1144.
23. Gan, W.; Sun, Y.; Sun, Y. Knowledge interaction enhanced knowledge tracing for learner performance prediction. In Proceedings of the 2020 7th International conference on behavioural and social computing (BESC). IEEE, 2020, pp. 1–6.
24. Zhou, Y.; Li, X.; Cao, Y.; Zhao, X.; Ye, Q.; Lv, J. LANA: towards personalized deep knowledge tracing through distinguishable interactive sequences. *arXiv preprint arXiv:2105.06266* **2021**.
25. Liu, S.; Yu, J.; Li, Q.; Liang, R.; Zhang, Y.; Shen, X.; Sun, J. Ability boosted knowledge tracing. *Information Sciences* **2022**, *596*, 567–587.
26. Sun, J.; Wei, M.; Feng, J.; Yu, F.; Li, Q.; Zou, R. Progressive knowledge tracing: Modeling learning process from abstract to concrete. *Expert Systems with Applications* **2024**, *238*, 122280.
27. Zhang, L. Learning factors knowledge tracing model based on dynamic cognitive diagnosis. *Mathematical Problems in Engineering* **2021**, *2021*, 8777160.
28. Zhu, J.; Yu, W.; Zheng, Z.; Huang, C.; Tang, Y.; Fung, G.P.C. Learning from interpretable analysis: Attention-based knowledge tracing. In Proceedings of the International conference on artificial intelligence in education. Springer, 2020, pp. 364–368.
29. Lee, U.; Park, Y.; Kim, Y.; Choi, S.; Kim, H. Monacobert: Monotonic attention based convbert for knowledge tracing. In Proceedings of the International Conference on Intelligent Tutoring Systems. Springer, 2024, pp. 107–123.
30. Zhang, M.; Zhu, X.; Zhang, C.; Qian, W.; Pan, F.; Zhao, H. Counterfactual Monotonic Knowledge Tracing for Assessing Students' Dynamic Mastery of Knowledge Concepts. In Proceedings of the Proceedings of the 32nd ACM international conference on information and knowledge management, 2023, pp. 3236–3246.
31. Karpatne, A.; Atluri, G.; Faghmous, J.; Steinbach, M.; Banerjee, A.; Ganguly, A.; Shekhar, S.; Samatova, N.; Kumar, V. Theory-guided Data Science: A New Paradigm for Scientific Discovery from Data. *IEEE Transactions on Knowledge and Data Engineering* **2017**, *29*, 2318–2331.
32. Karniadakis, G.E.; Kevrekidis, I.G.; Lu, L.; Perdikaris, P.; Wang, S.; Yang, L. Physics-informed machine learning. *Nature Reviews Physics* **2021**, *3*, 422–440.
33. Jia, X.; Willard, J.; Karpatne, A.; Read, J.S.; Zwart, J.A.; Steinbach, M.; Kumar, V. Physics-guided machine learning for scientific discovery: An application in simulating lake temperature profiles. *ACM/IMS Transactions on Data Science* **2021**, *2*, 1–26.
34. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational physics* **2019**, *378*, 686–707.
35. Willard, J.; Jia, X.; Xu, S.; Steinbach, M.; Kumar, V. Integrating scientific knowledge with machine learning for engineering and environmental systems. *ACM Computing Surveys* **2022**, *55*, 1–37.
36. Liu, Z.; Liu, Q.; Chen, J.; Huang, S.; Tang, J.; Luo, W. pyKT: a python library to benchmark deep learning based knowledge tracing models. *Advances in Neural Information Processing Systems* **2022**, *35*, 18542–18555.
37. Feng, M.; Heffernan, N.; Koedinger, K. Addressing the assessment challenge with an online system that tutors as it assesses. *User modeling and user-adapted interaction* **2009**, *19*, 243–266.
38. Stamper, J.; Niculescu-Mizil, A.; Ritter, S.; Gordon, G.; Koedinger, K. Algebra I 2005–2006 and bridge to algebra 2006–2007. *Development data sets from KDD Cup* **2010**.
39. Wang, Z.; Lamb, A.; Saveliev, E.; Cameron, P.; Zaykov, Y.; Hernández-Lobato, J.M.; Turner, R.E.; Baraniuk, R.G.; Barton, C.; Jones, S.P.; et al. Instructions and guide for diagnostic questions: The neurips 2020 education challenge. *arXiv preprint arXiv:2007.12061* **2020**.
40. Hewitt, J.; Liang, P. Designing and Interpreting Probes with Control Tasks. In Proceedings of the Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP); Inui, K.; Jiang, J.; Ng, V.; Wan, X., Eds., Hong Kong, China, 2019; pp. 2733–2743.
41. Belinkov, Y. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics* **2022**, *48*, 207–219.

**Disclaimer/Publisher's Note:** The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.